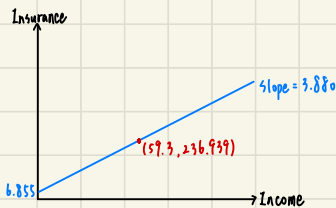


3.18

(a)

$$\text{Insurance} = 6.855 + 3.880 \cdot \text{Income} \\ (7.383) \quad (0.112)$$

$$\overline{\text{Insurance}} = 6.855 + 3.880 \cdot 59.3 = 236.939$$



(b)

$$N=20 \Rightarrow df=18 \Rightarrow t_{\frac{\alpha}{2}, 18} = 2.101$$

$$3.880 - 2.101 \cdot 0.112 = 3.6447$$

$$3.880 + 2.101 \cdot 0.112 = 4.1153$$

Point estimate = 3.880 $\Rightarrow$ For every \$1000 increase in Income, the amount of Insurance increases by \$3880

95% interval estimate = [3.6447, 4.1153]

In repeated sampling, 95% of intervals can capture the true value of β_2

(c)

$$\hat{\beta}_0 = 6.855 + 3.880 \cdot 100 = 394.855$$

$$\text{Var}(\hat{\beta}_0) = \text{Var}(\beta_0 + \beta_2 \cdot 100) = \sigma_{\beta_0}^2 + 10000 \sigma_{\beta_2}^2 + 2 \cdot 100 \cdot \sigma_{\beta_0, \beta_2} = 7.383^2 + 10000 \cdot 0.112^2 + 200 \cdot -0.746 = 30.7487$$

$$\Rightarrow SE(\hat{\beta}_0) = \sqrt{30.7487} = 5.5452$$

$$t_c = t_{\frac{\alpha}{2}, 18} = 2.878$$

$$394.855 - 2.878 \cdot 5.5452 = 378.896$$

$$394.855 + 2.878 \cdot 5.5452 = 410.814$$

99% interval estimate = [378.896, 410.814]

(d)

$$H_0: \beta_2 = 5$$

$$H_A: \beta_2 \neq 5$$

$$t^* = \frac{3.880 - 5}{0.112} = -10$$

$$RR = \{t: |t| > t_{\frac{\alpha}{2}, 18} = 2.101\}$$

$\therefore t^* \in RR \therefore$ We reject H_0

(e)

$$H_0: \beta_2 = 1$$

$$H_A: \beta_2 > 1$$

$$t^* = \frac{3.880 - 1}{0.112} = 25.714$$

$$RR = \{t: t > t_{\alpha, 18} = 2.552\}$$

$\therefore t^* \in RR \therefore$ We reject H_0

3.23

(a)

b2	0.184519002932086
df	498L
res_a	Named num [1:7] 20 7.38 0.21 -26.73 1.65 ...
res_b	Named num [1:7] 40 14.76 0.42 4.19 1.65 ...
se_b2	0.00525580090050928
t_crit	1.64791913885501

$$\text{Price} = \alpha_1 + \alpha_2 \text{sqft}^2 \Rightarrow ME(\text{sqft}) = \frac{d\text{Price}}{d\text{sqft}} = 2\alpha_2 \text{sqft}$$

$$\begin{cases} H_0: ME(20) \leq 13 \\ H_A: ME(20) > 13 \end{cases} \equiv \begin{cases} H_0: 40\alpha_2 \leq 13 \\ H_A: 40\alpha_2 > 13 \end{cases} \equiv \begin{cases} H_0: \alpha_2 \leq 0.325 \\ H_A: \alpha_2 > 0.325 \end{cases}$$

$$\widehat{ME(20)} = 40 \cdot 0.184519 = 7.38076$$

$$SE(\widehat{ME(20)}) = 40 \cdot SE(\alpha_2) = 40 \cdot 0.005256 = 0.21024$$

$$t^* = \frac{7.38076 - 13}{0.21024} = -26.73$$

$$RR = \{t \mid t_{0.95, 498} = 1.6473\}$$

$\therefore t^* \notin RR$ $\therefore$ We don't reject H_0 . Hence, the marginal effect on price of increasing the size of a 2000 square foot house is less than \$13,000.

P-value = 1

(b)

$$\begin{cases} H_0: ME(40) \leq 13 \\ H_A: ME(40) > 13 \end{cases} \equiv \begin{cases} H_0: 80\alpha_2 \leq 13 \\ H_A: 80\alpha_2 > 13 \end{cases} \equiv \begin{cases} H_0: \alpha_2 \leq 0.1625 \\ H_A: \alpha_2 > 0.1625 \end{cases}$$

$$\widehat{ME(40)} = 2 \cdot 40 \cdot 0.184519 = 14.76152$$

$$SE(\widehat{ME(40)}) = 80 \cdot SE(\alpha_2) = 80 \cdot 0.005256 = 0.42048$$

$$t^* = \frac{14.76152 - 13}{0.42048} = 4.1895$$

$$RR = \{t \mid t_{0.95, 498} = 1.6478\}$$

$\therefore t^* \in RR$ $\therefore$ We reject H_0 . Hence, the marginal effect on price of increasing the size of a 4000 square foot house is greater than \$13,000.

$$P(T_{498} > t^*) \approx 0.0000165$$

(c)

$$\widehat{E(\text{Price} | \text{sqft} = 20)} = 93.56585 + 0.184519 \times 20^2 = 167.37345$$

```
> predict(mod, newdata = data.frame(sqft = 20),
+         interval = "confidence", level = 0.95)
      fit      lwr      upr
1 167.3735 158.0481 176.6988
```

We estimate that houses with about 2000 square feet of living area sell for around \$167,373.5 on average. A 95% C.I. [158.0481, 176.6988] means that, in repeated sampling, 95% intervals can capture the true value of

(d)

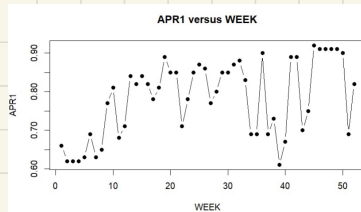
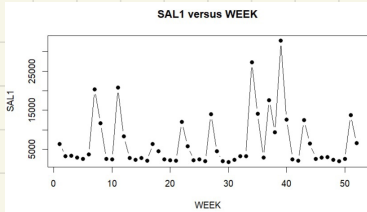
```
> # (d)
> mean(collegetown$price[collegetown$sqft == 20], na.rm = TRUE)
[1] 163.3333

> collegetown[collegetown$sqft == 20, ]
  price sqft age pool Fireplace close twostory occupied
121  138   20  8    0         0      0          0      0
188  169   20  8    0         1      0          0      0
262  183   20  7    0         1      0          0      1
```

$163.33 \in \text{C.I.}$, so the sample mean selling price of houses with $\text{sqft} = 20$ is compatible with the result in (c)

3.31

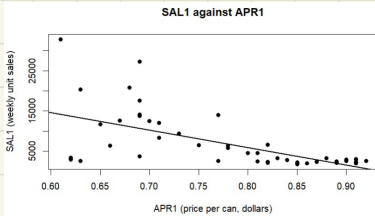
(a)



Variable	Mean	Min	Max	SD
SAL1	6718.7115	1762.00	32820.00	6.915083e+03
APR1	0.7825	0.61	0.92	9.803711e-02

The variation in sales is extremely high while price shows relatively low variation

(b)



Price and sales should generally have an inverse relationship. Therefore, if the scatterplot slopes downward from left to right, it is consistent with economic intuition.

(c)

```
> cat("b2_hat =", b2, "\n")
b2_hat = -434.4473
> cat("Interpretation of b2_hat: one-cent increase in price changes sales by", b2, "units.\n")
Interpretation of b2_hat: one-cent increase in price changes sales by -434.4473 units.
> cat("95% CI for b2 = ('", ci_b2_low, " ", ci_b2_high, ")\n\n")
95% CI for b2 = ( -592.2897 , -276.6049 )
```

After converting price into cents, B2 can be interpreted directly as the average change in sales associated with a one-cent increase in price.

(d)

```
> pred_70 <- predict(mod,
+                     newdata = data.frame(price1 = 70),
+                     interval = "confidence",
+                     level = 0.90)
>
> print(pred_70)
      fit      lwr      upr
1 10302.9 8624.934 11980.87
```

(e)

Elasticity = $\beta_2 \frac{\text{PRICE}_1}{\text{SAL}_1} \Rightarrow \widehat{\text{Elasticity}} = \frac{\widehat{\text{PRICE}_1}}{\widehat{\text{SAL}_1}} \cdot \widehat{\text{E}(\beta_2)}$

```
> cat("Mean(price1) = ", mean_price1, "\n")
Mean(price1) = 78.25
> cat("Mean(sal1) = ", mean_sal1, "\n")
Mean(sal1) = 6718.712
> cat("Elasticity at means = ", elas_hat, "\n")
Elasticity at means = -5.059825
> cat("SE(elasticity) = ", se_elas, "\n")
SE(elasticity) = 0.9152451
> cat("95% CI for elasticity = ('", ci_elas_low, " ", ci_elas_high, ")\n\n")
95% CI for elasticity = ( -6.898149 , -3.221501 )
```

The estimated price elasticity of demand at the means is -5.059825. This implies that, at the sample means, a 1% increase in the price of brand no. 1 tuna is associated with about a 5.06% decrease in unit sales. Since the absolute value of elasticity is greater than 1, demand is price elastic, indicating that sales are highly responsive to price changes.

(f)

```
> cat("H0: elasticity = -3\n")
H0: elasticity = -3
> cat("H1: elasticity != -3\n")
H1: elasticity != -3
> cat("t statistic = ", t_stat_f, "\n")
t statistic = -2.250572
> cat("Critical values = +/- ", t_crit_10_two_tail, "\n")
Critical values = +/- 1.675905
> cat("p-value = ", p_value_f, "\n")
p-value = 0.02884204
>
> if (abs(t_stat_f) > t_crit_10_two_tail) {
+   cat("Decision: Reject H0 at 10% significance level.\n")
+ } else {
+   cat("Decision: Do not reject H0 at 10% significance level.\n")
+ }
Decision: Reject H0 at 10% significance level.
```